Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
(Deemed to be University Estd. u/s 3 of UGC Act, 1956)

WEEKLY LESSON PLAN

Department: Computer Science and Engineering

Year / Semester: 2025-26 / Winter

Course Code / Course Name: 10212CS292 / Applied Programming Skills

Unit I: Linear Data Structures for Competitive Input Processing **Course Outcome:** Use linear data structure techniques to process competitive programming inputs efficiently.				
Week	**Session**	**Topics to be Covered**	**Problems to Practice (Hands-On)**	**Platform**
Week 1	Session 1	Course Orientation Time Complexity Analysis Two Pointers, Sliding Window	Task 1: **Squares of a Sorted Array** Task 2: **Contains Duplicate II**	Leet Code
	Session 2	Prefix Sum Difference Array	Task 3: **Running Sum of 1d Array** Task 4: **Sum of Absolute Differences in a Sorted Array**	
	Session 3	Binary Search Sorting	Task 5: **Binary Search** Task 6: **Sort Colors**	
	Session 4	Greedy pairing Hashing for frequency/count	Task 7: **Best Time to Buy and Sell Stock** Task 8: **First Unique Character in a String**	
	Session 5	In-place array manipulation	Task 9: **Move Zeroes** Task 10: **Reverse String**	

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Week 2	Session 6	Linked list fast/slow pointer	Task 11: **Middle of the Linked List** Task 12: **Palindrome Linked List**	Leet Code
	Session 7	Merge two lists	Task 13: **Merge Two Sorted Lists** Task 14: **Merge k Sorted Lists**	
	Session 8	Reverse nodes in k-group	Task 15: **Reverse Linked List** Task 16: **Reverse Nodes in k-Group**	
	Session 9	Detect/remove cycle	Task 17: **Linked List Cycle II**	
	Session 10	Intersection of linked lists	Task 18: **Intersection of Two Linked Lists**	

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Unit II: Stack and Queue Based Competitive Problem Modeling
Course Outcome: Demonstrate stack and queue operations to solve expression processing, scheduling, and data-handling problems.

Week	Session	Topics to be Covered	Problems to Practice (Hands-On)	Platform
Week 3	Session 11	Monotonic Stack - Next Greater Element pattern	Task 19: **Next Greater Element I** Task 20: **Daily Temperatures**	Leet Code
	Session 12	Stock Span pattern	Task 21: **Online Stock Span** Task 22: **Final Prices With a Special Discount in a Shop**	
	Session 13	Valid parentheses variants	Task 23: **Valid Parentheses** Task 24: **Minimum Remove to Make Valid Parentheses**	
	Session 14	Min Stack design	Task 25: **Min Stack**	
	Session 15	Stack simulation problems	Task 26: **Asteroid Collision** Task 27: **Validate Stack Sequences**	

Week	Session	Topics to be Covered	Problems to Practice (Hands-On)	Platform
Week 4	Session 16	Queue using stacks	Task 28: **Implement Queue using Stacks**	Leet Code
	Session 17	Sliding window maximum using deque	Task 29: **Sliding Window Maximum** Task 30: **Longest Continuous Subarray With Absolute Diff Less Than or Equal to Limit**	
	Session 18	BFS-level processing with queue	Task 31: **Binary Tree Level Order Traversal** Task 32: **Binary Tree Right Side View**	
	Session 19	Circular queue simulation	Task 33: **Design Circular Queue** Task 34: **Design Circular Deque**	
	Session 20	Task scheduling with queue	Task 35: **Task Scheduler** Task 36: **Number of Recent Calls**	